



Universidade do Minho
Escola de Ciências

Departamento de Matemática



Cálculo para Engenharia

Licenciatura em Engenharia Informática

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Algumas regras de derivação

(estão a ser omitidos os domínios das funções)

$$C' = 0, \quad C \text{ constante}$$

$$(fg)'(x) = f'(x)g(x) + f(x)g'(x)$$

$$(g \circ f)'(x) = g'(f(x))f'(x)$$

$$(e^x)' = e^x$$

$$(a^x)' = a^x \ln a$$

$$\operatorname{sen}'x = \cos x$$

$$\operatorname{tg}'x = \sec^2 x$$

$$\sec'x = \sec x \operatorname{tg} x$$

$$\operatorname{sh}'x = \operatorname{ch} x$$

$$\operatorname{th}'x = \operatorname{sech}^2 x$$

$$\operatorname{sech}'x = -\operatorname{sech} x \operatorname{th} x$$

$$\operatorname{arcsen}'x = \frac{1}{\sqrt{1-x^2}}$$

$$\operatorname{arctg}'x = \frac{1}{1+x^2}$$

$$\operatorname{arcsec}'x = \frac{1}{x\sqrt{x^2-1}}$$

$$\operatorname{argsh}'x = \frac{1}{\sqrt{1+x^2}}$$

$$\operatorname{argth}'x = \frac{1}{1-x^2}$$

$$\operatorname{argsech}'x = \frac{-1}{x\sqrt{1-x^2}}$$

$$(x^\alpha)' = \alpha x^{\alpha-1}, \quad (\alpha \in \mathbb{R})$$

$$\left(\frac{f}{g}\right)'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{g^2(x)}$$

$$(f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))}$$

$$\ln'x = \frac{1}{x}$$

$$\log'_a x = \frac{1}{x \ln a}$$

$$\cos'x = -\operatorname{sen} x$$

$$\cotg'x = -\operatorname{cosec}^2 x$$

$$\operatorname{cosec}'x = -\operatorname{cosec} x \cotg x$$

$$\operatorname{ch}'x = \operatorname{sh} x$$

$$\operatorname{coth}'x = -\operatorname{cosech}^2 x$$

$$\operatorname{cosech}'x = -\operatorname{cosech} x \operatorname{coth} x$$

$$\operatorname{arccos}'x = \frac{-1}{\sqrt{1-x^2}}$$

$$\operatorname{arccotg}'x = \frac{-1}{1+x^2}$$

$$\operatorname{arcosec}'x = \frac{-1}{x\sqrt{x^2-1}}$$

$$\operatorname{argch}'x = \frac{1}{\sqrt{x^2-1}}$$

$$\operatorname{argcoth}'x = \frac{1}{1-x^2}$$

$$\operatorname{argcosech}'x = \frac{-1}{x\sqrt{1+x^2}}$$